A-A-50755
15 October 1986

COMMERCIAL ITEM DESCRIPTION

CONTROL HEAD ASSEMBLY TEST FIXTURE

The General Services Admininstration has authorized the use of this commercial item description.

SCOPE. This Commercial Item Description (CID) specifies a test fixture that test the control head assembly P/N 627-5023-001 by providing a drive current to both the azimuth and elevation servo motors, by supplying 28VDC power to four modules within the head assembly, and by receiving and monitoring the control head assembly load. This system shall be known as a control head assembly test fixture.

APPLICABLE DOCUMENTS.

SPECIFICATIONS

MILITARY

MIL-M-7298	Manual Technical, Commercial Equipment.
MIL-T-28800	Test Equipment for use with Electrical and Electronic Equipment, General Specification for.
MIL-M-38797	Manual Technical, Operation Instructions and Maintenance Instructions (for various types of Equipment).
MIL-C-38999	Connector, Receptacle, Electric, Wall Mounting Flange, Crimp Type, Bayonet Coupling, Series II.

STANDARDS

MILITARY

MIL-STD-12C	Abbreviations for Use on Drawings, Specifications, Standards and in Technical Documents.
MIL-STD-15-1A	Electrical and Electronics Symbols.
MIL-STD-681B	Identification Coding and Application of Hook Up and Lead Wire.
MIL-STD-810D	Environmental Test Methods and Engineering Guidelines.

OTHER PUBLICATIONS

USAS Y32.16-1965	Electrical and Electronics Reference Designations.
T.O. 00-25-234	General Shop Practice Requirements for the Repair, Maintenance and Test of Electronic Equipment.

SALIENT CHARACTERISTICS.

1. Performance requirements.

   a. Control head assembly test fixture. The control head assembly test fixture has an external power supply of 28 ±1.0VDC, 16.07 AMPS

FSC 6625

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maximum and is controlled by a relay operated switch on the test fixture. The relay switch causes illumination of a light indicator, and delivers 28VDC power to four modules in the Unit Under Test (UUT) and the azimuth and elevation of the control head assembly. A slo-blo fuse for the 28VDC input will be used to protect the test fixture and the UUT. The test fixture can also receive and monitor the UUT output signals.

b. Receiving and monitoring of UUT output signals. The UUT output signals will load into the test fixture to an azimuth current sink and an elevation current sink. The current sinks are made up of two parallel connected 8.66 OHM load resistors at 120 watts. There will be three pairs of voltage monitoring tip jacks (isolated) which will be measured from a digital multimeter. One pair is used to monitor the input voltage and the other two are used to monitor the UUT output signals. The UUT output signals should be measured in parallel with the resistors. A thermal protection switch will remove power from the test fixture if the heat sinks temperature raises above 203 degrees Fahrenheit. A cooling fan will be used to protect the azimuth and elevation heat sink load resistors. The test fixture will supply a grounding clip, 12 inches from the front panel, to assure UUT grounding capability.

c. Azimuth and elevation drive controls. Two toggle switches one for elevation control and another for azimuth control signals are located at the front panel. The elevation switch should indicate up and down for vertical antenna control and the azimuth switch should indicate CW or CCW for horizontal antenna control. The control signals are 28VDC output and are supplied to the UUT through the connector.

d. Plug in connector. The plug in connector military standard MIL-C-38999, P/N MS27472T14B-18S, 359-0689-010, has a 12 inch cable from the front panel of the test fixture, and its pin locations as shown in figure 1-1 and pin assignment as shown in table 1-1. The four module inputs A1, A2, A3, A4, are in pins N, R, M, L, E, G, D, C. The antenna control drives are in pins P, S, F, H, and the UUT signal inputs are in pins A, B, J, K.

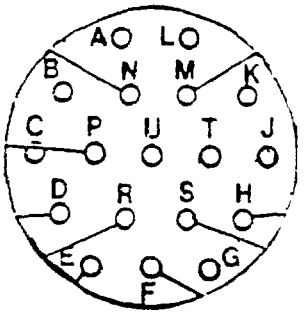


Figure 1-1. Test fixture test point locations.

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e. Operating temperature. The operation temperature of the test fixture is 50.0 to 104.0 degrees F.

f. Storage temperature. The storage temperature of the test fixture is -67.0 to 167.0 degrees F.

g. Humidity and fungus. The test fixture humidity is 95(+5,-0)% limited to 75% above 77 degrees F and the fungus resistance is according to MIL-STD-810, method 508.

h. Vibration and shock. The test fixture must handle 2 x G force at 55 Hz maximum vibration and a shock of 30 x G force at 18 shocks.

2. Type and size.

a. Design and construction. Unless otherwise specified herein, the equipment shall comply with the type III design and construction requirements specified in MIL-T-28800.

b. Solid state construction. The equipment shall be of solid-state modular miniaturized construction.

Table 1-1. Connector pin assignment chart.

Connector	Pin	Function
MS27472T14B-18S	A	Azimuth motor load from modules A4 and A3 +.
	B	Azimuth motor load from modules A4 and A3 -
	C	+28VDC return for A4 module (GND).
	D	+28VDC to A4 module.
	E	+28VDC to A3 module.
	F	+28VDC control input for A4 antenna control in the clockwise direction.
	G	+28VDC return for A3 (GND).
	H	+28VDC control input for A3 antenna control in the counter clockwise direction.
	J	Elevation motor load from modules A1 and A2 +.
	K	Elevation motor load from modules A1 and A2 -.
	L	+28VDC to A2 module (GND).
	M	+28VDC to A2 module.
	N	+28VDC to A1 module.
	P	+28VDC control input for A2 antenna control in the upward direction.
	R	+28VDC return for A1 module (GND).
	S	+28VDC control input for A1 antenna control in the downward direction.

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c. Receptacle and monitor connections. All connections are coded in red for positive, black for negative, and green for ground. They will have plastic insulation, the current carrying parts and springs will be gold-plated brass for conductivity and corrosion resistance, and they will be open binding post to accept a variety of connections. The power supply connects to the test fixture through receptacle connections that has a wire hole diameter of .098 ±.002. The connections should be equal to P/N BP30GN Superior Electronic Co. The digital multimeter connects to the test fixture through the monitor connections that has a wire hole diameter of .080 ±.001. The connection should be equal to P/N TJ100 Ratheon Co.

d. Accessibility. The equipment shall be constructed so subassemblies and chassis components can be removed without removing any other hardware subassembly, printed circuit card or component.

e. Input power switch. The input power switch shall comply with type III input switch requirements of MIL-T-28800, except that the switch shall break both sides of the power source, but not break the earth ground conductor circuit.

f. Fuses or circuit breakers. Fuses or circuit breakers shall comply with type III requirements of MIL-T-28800, except that both sides of the input power line shall be equipped with fuses or circuit breakers.

3. Dimensions, weight and mechanical stability.

a. Weight. The unpacked weight of the test fixture, including all accessories, shall not exceed 15.0 pounds.

b. Dimensions. The dimensions of the test fixture shall be 8.5 inches in height, 14.0 inches in width and 12.0 inches in length nominal.

CERTIFICATION. The contractor shall certify that the product offered meets the salient characteristics of this description, and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices. Bid samples from bidders will be tested in accordance with the salient characteristics of the Commercial Item Description at San Antonio ALC, Kelly AFB, TX 78241.

OPERATIONAL INSTRUCTIONS/MAINTENANCE MANUAL. The contractor shall prepare an operator's manual according to the provisions of MIL-M-38797 for all equipment delivered under terms of this contract.

COMMERCIAL PUBLICATIONS. Commercial publications may be submitted to fulfill Air Force documentation requirements. These publications require technical order numbering and must be referenced in the system level publication. Such commercial data must be prepared in accordance with MIL-M-7298C. Two copies of each commercial manual proposed shall be submitted for Air Force evaluation and approval. Commercial publications shall be reviewed by the Technical Order Management Office (TOMO) before final acceptance by the Air Force. If modifications are performed on Air Force equipment, the contractor shall provide updating manual changes to meet Air Force standards. All applicable reproduction rights for government use must accompany submittals.

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REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical.

PACKAGING, PACKING AND MARKING. The items shall be packaged in accordance with normal commercial practice and packed to assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling, and storage. The shipping container shall be in compliance with national motor freight classification and uniform freight classification. Shipping containers shall be marked as specified in the contract or order.

KNOWN ACCEPTABLE PRODUCTS.

Manufacturer	Model	NSN
Rockwell International Corp For Competitive Bidding	P/N 622-5844-001	6625-01-103-1226

Custodian.
Navy - EC
Air Force - 99

Review activity:
DLA - ES

Preparing activity:
Air Force - 82

(Project 6625-0658)

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